ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** chemical reaction	Law.
2	**(d)** cation	Cation.
3	**(d)** ionic	Transfer.
4	**(c)** Lavoisier	Lavoisier.
5	**(d)** anion	Anion.
6	**(a)** covalent	Sharing.
7	**(d)** 1 : 8	By mass.
8	**(b)** Na^+	Loses 1 electron.
9	**(d)** solids with high melting point	Crystalline.
10	**(a)** matter	Matter.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass is neither created nor destroyed in a chemical reaction (2).	2
12	A charged atom or group of atoms (2).	2
13	Ionic bond (2).	2
14	A compound has a fixed ratio of elements by mass (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Matter has atoms (1); Atoms of an element are identical (1); Atoms combine in whole-number ratios (1).	1+1+1
16	Low melting point (1); Do not conduct (1); Soluble in organic solvents (1).	1+1+1
17	Gain electrons (1); Cl^- (1); O^{2-} (1).	1+1+1
	OR: 12 + 32 (1); 44 g (1); Law of conservation of mass (1).	1+1+1
18	1, 3, 4 (1); 8, 2 (1); 5 (1).	1+1+1
	OR: Na^+ (1); Ca^{2+} (1); Al^{3+} (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Statement (1); Reactants and products (1); Burning of carbon (1); 12 g + 32 g = 44 g (1); Closed system (1).	1+1+1+1+1
	OR: NaCl (1); MgO (1); CaCl2 (1); Al2O3 (1); H2O (1).	1+1+1+1+1
20	Statement (1); Water ratio (1); 1 : 8 (1); Any source of water (1); Not for mixtures (1).	1+1+1+1+1
	OR: H2O 18 (1); CO2 44 (1); NH3 17 (1); CH4 16 (1); HCl 36.5 (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.
- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Cl^-	Gains 1 electron.
2	**(c)** low melting points	Weak forces.
3	**(d)** atomic theory	Atomic theory.
4	**(b)** O^{2-}	Gains 2 electrons.
5	**(c)** u	Unified atomic mass unit.
6	**(b)** 2 atoms	O2.
7	**(a)** $NaCl$	1:1.
8	**(b)** 12 u	C = 12.
9	**(d)** 3 atoms	O3.
10	**(b)** MgO	2+ and 2-.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Positively charged ion (1); Na^+ (1).	1+1
12	H_2O (1); CO_2 (1).	1+1
13	Group of atoms chemically bonded that can exist independently (2).	2
14	Negatively charged ion (1); Cl^- (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	NH_3 = 17 u (1); CH_4 = 16 u (1); N = 14, H = 1, C = 12 (1).	1+1+1
16	Water H_2O (1); Ammonia NH_3 (1); Methane CH_4 (1).	1+1+1
17	Cl^- (1); O^{2-} (1); NO_3^- (1).	1+1+1
	OR: CO_2 = 44 u (1); SO_2 = 64 u (1); O = 16 (1).	1+1+1
18	H_2 (1); O_2 (1); N_2 (1).	1+1+1
	OR: Atom is one (1); Molecule is a group (1); Molecule is formed by bonds (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Atoms (1); Identical atoms (1); Different elements (1); Combine (1); Not created (1).	1+1+1+1+1
	OR: Definition (1); Na+ (1); Mg2+ (1); Cl- (1); O2- (1).	1+1+1+1+1
20	Sodium loses an electron (1); Chlorine gains it (1); Na+ and Cl- (1); Attraction (1); Ionic bond (1).	1+1+1+1+1
	OR: H2: 2 (1); O3: 3 (1); P4: 4 (1); S8: 8 (1); He: 1 (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 16 u	O = 16.
2	**(d)** 3 atoms	H2O.
3	**(d)** CaO	2+ and 2-.
4	**(b)** 32 u	2 x 16.
5	**(d)** CO_2	CO2.
6	**(d)** HCl	1:1.
7	**(c)** 2 u	2 x 1.
8	**(d)** NH_3	NH3.
9	**(d)** H_2S	H+ and S2-.
10	**(a)** 44 u	12 + 32.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2 x 1 + 16 (1); = 18 u (1).	1+1
12	Number of atoms in a molecule (1); 2 (1).	1+1
13	Covalent bond (2).	2
14	23 + 35.5 (1); = 58.5 u (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Na_2O (1); $MgCl_2$ (1); CaO (1).	1+1+1
16	H_2O (1); CO_2 (1); NH_3 (1).	1+1+1
17	1 : 8 (1); 16 g (1); Hydrogen 2 g (1).	1+1+1
	OR: Solids (1); High melting point (1); Conduct when molten (1).	1+1+1
18	Lose electrons (1); Na^+ (1); Mg^{2+} (1).	1+1+1
	OR: $NaCl$ (1); MgO (1); $CaCl_2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Each H has one electron (1); They share (1); One pair (1); Covalent bond (1); H2 formed (1).	1+1+1+1+1
	OR: Solids (1); High melting (1); Brittle (1); Conduct in solution (1); Dissolve in water (1).	1+1+1+1+1
20	Transfer (1); Sharing (1); Metals and non-metals (1); Non-metals (1); Examples (1).	1+1+1+1+1
	OR: Low melting (1); Poor conductors (1); Soft (1); Soluble in organic solvents (1); Many are gases or liquids (1).	1+1+1+1+1

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